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Solving Linear Equations: The Nonhomogeneous Case

A system $Ax = b$ where $b \neq 0$ is said to be non-homogeneous. For a nonhomogeneous system, we introduce the augmented coefficient matrix $[A \vdots b]$.

Fact: If a system with augmented coefficient matrix $[A \vdots b]$ has a particular solution x_p, then any solution of the nonhomogeneous system can be represented by $x_p + x_h$ where x_h is the solution of the homogeneous system.
Therefore, the procedure of finding the general solution of a nonhomogeneous linear equation can be summarized as

1) Construct the augmented coefficient matrix $[A \vdots b]$.
2) Find the reduced echelon form of the augmented coefficient matrix.
3) Using the first part of the reduced echelon form, find the solution of the homogeneous system $Ax_h = 0$.
4) Using the complete reduced echelon form, find a particular solution x_p.
5) Express the general solution by $x = x_h + x_p$.

Solving Linear Equations: The Nonhomogeneous Case

Example

Find the solution set of the linear system

$$x_1 + 2x_2 - x_4 = 1$$

$$-2x_1 - 3x_2 + 4x_3 + 5x_4 = 2$$

$$2x_1 + 4x_2 - x_4 = 4$$

Solving Linear Equations: The Nonhomogeneous Case

Example

Find the solution set of the linear system

$$x_1 + 2x_2 - x_4 = 1$$

$$-2x_1 - 3x_2 + 4x_3 + 5x_4 = 2$$

$$2x_1 + 4x_2 - x_4 = 4$$

Solution: First, we represent the linear system in $Ax = b$ form:

$$\begin{bmatrix} 1 & 2 & 0 & -1 \\ -2 & -3 & 4 & 5 \\ 2 & 4 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$$

Solving Linear Equations: The Nonhomogeneous Case

Step 1) Construct the augmented coefficient matrix $[A \vdots b]$.

$$\left[\begin{array}{ccccc} 1 & 2 & 0 & -1 & 1 \\ -2 & -3 & 4 & 5 & 2 \\ 2 & 4 & 0 & -1 & 4 \end{array}\right]$$

Step 2) Find the reduced echelon form of the augmented coefficient matrix.

$$\left[\begin{array}{ccccc} 1 & 0 & -8 & 0 & 7 \\ 0 & 1 & 4 & 0 & -2 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right] = [A_r \vdots b_r]$$

Solving Linear Equations: The Nonhomogeneous Case

Step 3) Using the first part of the reduced echelon form, find the solution of the homogeneous system $A_r x_h = 0$.

$$x_h = \left\{ s \begin{bmatrix} 8 \\ -4 \\ 1 \\ 0 \end{bmatrix}, \forall s \in \mathbb{R} \right\}$$

Step 4) Using the complete reduced echelon form, find a particular solution x_p. $x_1 = 8x_3 + 7$, $x_2 = -4x_3 - 2$ and $x_4 = 2$. Since x_3 is a free variable, we choose any value for x_3. For $x_3 = 0$, a particular solution will be $x_1 = 7$, $x_2 = -2$, $x_3 = 0$ and $x_4 = 2$.

$$x_p = \begin{bmatrix} 7 \\ -2 \\ 0 \\ 2 \end{bmatrix}$$

Solving Linear Equations: The Nonhomogeneous Case

Step 5) Express the general solution by $x = x_h + x_p$.

$$x = x_h + x_p = \left\{ s \begin{bmatrix} 8 \\ -4 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 7 \\ -2 \\ 0 \\ 2 \end{bmatrix}, \forall s \in \mathbb{R} \right\}$$

Solving Linear Equations: The Nonhomogeneous Case

Step 5) Express the general solution by $x = x_h + x_p$.

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Homework Exercise: Find the solution of the following equation set

$$x_1 + 3x_2 + 3x_3 = 1$$

$$2x_1 + 6x_2 + 9x_3 = 5$$

$$-x_1 - 3x_2 + 3x_3 = 5$$

Answer: $x = \left\{ \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + s \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}, \forall s \in \mathbb{R} \right\}$

Solving Linear Equations: The Nonhomogeneous Case

Homework Exercise: Find the solution of the following equation set

$$x_3 + 4x_4 = 6$$

$$x_1 + 3x_2 + 2x_4 = 1$$

$$x_1 + 3x_2 + x_3 + 6x_4 = 7$$

Answer: $x = \left\{ \begin{bmatrix} 1 \\ 0 \\ 6 \\ 0 \end{bmatrix} + s \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -2 \\ 0 \\ -4 \\ 1 \end{bmatrix}, \forall s, t \in \mathbb{R} \right\}$

Transpose of Vectors and Matrices

Transpose of a matrix is formed by turning all the rows of a given matrix into columns and vice-versa.
Transpose of a column vector is a row vector with same components and vice-versa.

Example

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad A^\top = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad x^\top = [1 \;\; 2 \;\; 3]$$

$$y = [4 \;\; 5 \;\; 6], \quad y^\top = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

Note that transpose of an $m \times n$ matrix is an $n \times m$ matrix.

Inverting a Matrix

Definition

(Identity matrix) The identity matrix is a square matrix which consists of ones on the diagonal and zeros elsewhere. The identity matrix of dimension n is denoted by I_n.

The inverse of an $n \times n$ matrix A, denoted by A^{-1}, is an $n \times n$ matrix which satisfies $AA^{-1} = A^{-1}A = I_n$.

There are various ways to find the inverse of a matrix. One of these ways is called 'Gaussian elimination' which is composed of the following steps:

1) Construct the augmented matrix $[A \ \vdots \ I_n]$.
2) Apply elementary row operations to have a matrix of the form $[I_n \ \vdots \ B]$.
3) B is the inverse of A.

Inverting a Matrix

Example

Find the inverse of $A = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}$.

Inverting a Matrix

Example

Find the inverse of $A = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}$.

Since the dimension of A is 3, we construct the augmented matrix $[A \vdots I_3]$ and apply elementary row operations as follows

$$[A \vdots I_3] = \begin{bmatrix} 1 & 2 & 0 & 1 & 0 & 0 \\ 1 & 3 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{bmatrix} \overset{R_2 - R_1}{\rightarrow} \begin{bmatrix} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{bmatrix} \overset{R_3 - R_1}{\rightarrow}$$

$$\begin{bmatrix} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{bmatrix} \overset{R_1 - 2R_2}{\rightarrow} \begin{bmatrix} 1 & 0 & -2 & 3 & -2 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{bmatrix} \overset{R_2 - R_3}{\rightarrow}$$

Inverting a Matrix

$$\begin{bmatrix} 1 & 0 & -2 & 3 & -2 & 0 \\ 0 & 1 & 0 & -2 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{bmatrix} \overset{R_1+2R_3}{\rightarrow} \begin{bmatrix} 1 & 0 & 0 & 5 & -4 & 2 \\ 0 & 1 & 0 & -2 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{bmatrix}.$$

The inverse of A is computed as $\begin{bmatrix} 5 & -4 & 2 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{bmatrix}$. The result can be verified by showing $AA^{-1} = I_3$ or $A^{-1}A = I_3$.

Homework Exercise: Find the inverse of $\begin{bmatrix} 0 & 1 & 2 \\ 3 & 0 & 4 \\ 2 & 6 & 4 \end{bmatrix}$ and check your answer by satisfying the equations $AA^{-1} = I_3$ and $A^{-1}A = I_3$.

Answer:

$$\begin{bmatrix} -3/4 & 1/4 & 1/8 \\ -1/8 & -1/8 & 3/16 \\ 9/16 & 1/16 & -3/32 \end{bmatrix}$$

Determinant of a Matrix

Determinant of an $n \times n$ matrix A, denoted by $det(A)$ or $|A|$, is a scalar function.

Determinant of a 1×1 Matrix:

The determinant of a 1×1 matrix is defined to be equal to itself, i.e., if $A = [a]$, then $det(A) = a$.

Determinant of a 2×2 Matrix:

Consider the 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Its determinant can be computed by $det(A) = ad - bc$.

Determinant of a Matrix

Determinant of a 3×3 Matrix:

Consider the 3×3 matrix $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$. Its determinant can be computed by $det(A) = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix} =$

$a(ei - fh) - b(di - fg) + c(dh - eg) = aei - afh - bdi + bfg + cdh - ceg.$

Eigenstructure of a Matrix

Definition (Eigenvalues of a Matrix)

The eigenvalues of an $n \times n$ matrix A are the roots of the equation

$$det(\lambda I_n - A) = 0 \quad \text{or} \quad det(A - \lambda I_n) = 0$$

This equation is called the characteristic equation of A.

Definition (Eigenvectors of a Matrix)

An eigenvector of an $n \times n$ matrix A, corresponding to the eigenvalue λ, is the vector v which satisfies the equation

$$Av = \lambda v$$

Note that eigenvectors are not unique. If v is an eigenvector corresponding to the eigenvalue λ, then kv is also an eigenvector corresponding to the eigenvalue λ where k is a nonzero scalar.

Eigenstructure of a Matrix

Example

Compute the eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$.

Eigenstructure of a Matrix

Example

Compute the eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$.

In order to compute the eigenvalues, we must find the roots of the equation

$$det(\lambda I_n - A) = 0 \quad \rightarrow \quad \begin{vmatrix} \lambda - 1 & -2 \\ -2 & \lambda - 1 \end{vmatrix} = (\lambda - 1)^2 - 4 = 0$$

$$(\lambda - 3)(\lambda + 1) = 0 \quad \rightarrow \quad \lambda_1 = 3, \quad \lambda_2 = -1$$

To find an eigenvector of A corresponding to the eigenvalue $\lambda_1 = 3$ is computed by

$$Av_1 = \lambda_1 v_1 \quad \rightarrow \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} = 3 \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} \quad \rightarrow \begin{bmatrix} v_{11} + 2v_{12} \\ 2v_{11} + v_{12} \end{bmatrix} = \begin{bmatrix} 3v_{11} \\ 3v_{12} \end{bmatrix}$$

$$v_{11} + 2v_{12} = 3v_{11} \quad \text{and } 2v_{11} + v_{12} = 3v_{12} \rightarrow v_1 = [1 \ 1]^\top$$

Eigenstructure of a Matrix

Homework Exercise: Find the eigenvalues and a set of corresponding eigenvectors of the matrix

$$\begin{bmatrix} 4 & -4 & -2 \\ 2 & -2 & -2 \\ 4 & -4 & -2 \end{bmatrix}$$

Answer: Eigenvalues are 2, -2 and 0. Corresponding eigenvectors are $[1 \;\; 0 \;\; 1]^\top$, $[1 \;\; 1 \;\; 1]^\top$ and $[1 \;\; 1 \;\; 0]^\top$, respectively.